

Lucas Zheng

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EDUCATION

University of California, Berkeley

Expected graduation: May 2026

Bachelor's, Computer Science

GPA: 4.0

Discrete Math and Probability Theory (A+), Data Structures (A+), Structure and Interpretation of Computer Programs (A+), Abstract Linear Algebra (A+), Sound and Music Computing (A+), Computer Architecture, Internet Architecture and Protocols, Probability and Random Processes, Numerical Analysis

DeAnza College

Intro Linux/Unix (A), Intro C++ (A+), Intermediate C++ (A)

AWARDS

ICPC: 19th place in the 2024 Inter-Collegiate Programming Contest (ICPC) Pacific NorthWest Regional.

USA Computing Olympiad: 2023 US Open Gold Division 154th place.

USA Math Olympiad: 4x American Invitational Mathematics Exam (AIME) qualifier, 2x AMC distinction (top 5%)

Math League: 2023 International Math League 5th place, 2x North California Regional 2nd place

XPRIZE: Avatar XPRIZE semifinalist, Wildfire XPRIZE top 30

PROJECTS

FlySafe: Inventor (<https://3lucasz.github.io/my-site/flysafe>)

- Invented an affordable airplane altimeter using ESP32, LIDAR, and WebSocket technology. The device helps amateur pilots improve their landing skills by delivering real-time visual and audio feedback integrated with the aircraft sound system.
- Developed an app for real-time data monitoring, device configuration, and post-flight data analysis, providing pilots with insights to improve their skills.

EZ-Check: Inventor (<https://3lucasz.github.io/my-site/ezcheck>)

- Invented a novel IoT authentication system to secure the machinery of a school's lab. Integrated electronics and backend systems, designed 3D enclosures, and created a custom PCB.
- Built a platform for administrators to manage lab users, supervisors, and machines, streamlining all operations of the maker lab.

EZ-Find: Inventor

- Created an innovative inventory management app used across a school's storages for efficient item tracking and locating.
- Developed a platform that simplifies the management of users, organizations, storages, and inventory. Integrated a QR-code system and automatic label printing functionality.

Google Tensorflow Lite for Microcontrollers Challenge

- Developed an interactive AI-powered pet by integrating BLE, IMU, and Tensorflow to give the pet a sense of emotion.
- Self-taught TinyML through EdX Harvard courses.

EXPERIENCE

Carnegie Mellon University: Undergraduate Researcher

August 2024 - Present

- Conducting research with Professor Theophilus Benson to publish a paper on Web Benchmarking.
- Read research papers on web complexities and web optimizations.
- Wrote scripts to efficiently extract and cache static, dynamic, and operational complexities from 30,000 websites using tools like Puppeteer, Chrome DevTools Protocol, and MemInfra.
- Used K-Means clustering to discover 22 clusters that represent a corpus of websites.

VCS Research and Development Fellow

August 2022 - August 2024

Underwater360 (<https://3lucasz.github.io/my-site/underwater360>)

- Developed a remotely controllable underwater 360° camera system to enhance marine biology education through immersive experiences.
- Hacked an Insta360 by integrating it with a Khadas VIM4 SBC and developing an Android-based server solution.

- Developed a desktop app for camera operation with interactive 360° live streaming and customizable capture settings.

MBARI AUV: Software Developer

- Contributed to the development of an autonomous underwater vehicle capable of delivering experiments up to 1 km below water.
- Implemented underwater localization using ROS2 in Gazebo simulator, integrating Kalman Filter sensor fusion with USBL beacon, IMU, and dynamics data.

Gimbal: Co-inventor

- Developed a controllable 3-axis auto-stabilizing gimbal by modifying a RoninSC Gimbal with a serial inverter and SBUS, developing a ROS node for robot integration, and streaming a stereoscopic camera to a VR headset.

Quest Institute RoboQuest: Program Developer

October 2019 - August 2022

Hackbot (<https://3lucasz.github.io/my-site/hackbot>)

- Enhanced program kit capabilities by developing software and a control framework to integrate 2D LIDAR mapping, immersive VR teleoperation, and Tensorflow.
- Developed software for a Raspberry Pi tank to autonomously navigate a track with Tensorflow, competing in a Donkey Car Event.

Other responsibilities:

- Built a robot using Jetson Nano and ROS2 to explore potential for a new school program and developed a website for remote teleoperation and live camera streaming.
- Assisted in the development and testing of program curriculum and robots. Taught students to build and program robots using micro:bit through virtual classes during COVID19.

Avatar XPRIZE Team Avatar Quest: Software Developer

August 2021 - September 2021

- Qualified for the semifinals of the competition. We developed a humanoid robot with telepresence capabilities.
- Contributed to the development of software for wrist control and gripper functions.

Wildfire XPRIZE Team Wildfire Quest: Lead / Advisor

July 2022 - July 2023 / August 2023 - August 2024

- Qualified for Track B (top 30), winning \$20,000. We designed a system for the rapid detection and suppression of wildfires over a large area. Appointed to lead a team of 9 at the inception of the project, overseeing all aspects of project development and delivery of the technical submission.

FRC Robotics Team 3256: Software Developer / Software Co-lead

October 2020 - June 2023 / June 2023 - May 2024

- Qualified for the World Championship in 2022, 2023, and 2024.
- Developed robotics solutions including dynamic path generation with Bezier curves and Dijkstra's algorithm, LED animation library, match scouting app, and programming/simulation of robotic subsystems.
- Led training sessions by delivering lectures on complex topics, reviewing code commits, and mentoring peers to elevate team performance.